SARATH RADHAKRISHNAN

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PROFESSIONAL SUMMARY

Computational engineer with 7 years of experience in CFD, Large-Eddy Simulation (LES), non-linear flow physics, and scientific computing. Developed a data-driven wall model for LES of turbulent wall-bounded flows during doctoral research, combining turbulence modeling with machine-learning methods. Brings aerospace propulsion experience from SCRAMJET systems engineering at ISRO to large-scale CFD and scientific computing research. Contributed to reduced-order modeling by generating and analysing high-fidelity urban flow simulation datasets. Mentored graduate students and delivered science outreach.

EDUCATION

Universitat Politecnica de Catalunya

Sep 2018 - Jun 2024 | *Barcelona, Spain*

PhD

Thesis: Non-Equilibrium Wall-modeling in Large Eddy Simulation of High-speed Transitional Flows. Advisors: Oriol Lehmkuhl and Daniel Mira.

Ecole Centrale de Nantes

Sep 2015 - Aug 2017 | *France*

Master of Engineering in Computational Mechanics

Thesis: Modelling of Uterine Synchronisation by a Hydro-Mechanical and Electrical Activation Mechanism — Mechanical Modelling of Uterus. Advanced coursework in continuum mechanics, numerical methods, finite element methods, and computational modelling. Internship project funded by ANSYS; used ANSYS-APDL to model and simulate uterine synchronization.

National Institute of Technology Calicut

Jul 2004 - Jun 2008 | *India*

Bachelor of Technology in Mechanical Engineering

RESEARCH INTERESTS

- Wall-modeled Large-Eddy Simulation (LES) of turbulent wall-bounded flows
- Non-equilibrium wall modeling, transitional flows, and relaminarization
- Data-driven and physics-informed modeling for CFD (Scientific Machine Learning)
- Reduced-order and surrogate modeling for engineering applications
- High-performance computing and large-scale numerical simulation
- Aerospace fluid dynamics and propulsion (compressible, aero-thermal flows)
- Urban environmental fluid mechanics and digital twins

WORK EXPERIENCE

Barcelona Supercomputing Center

Aug 2024 – Jun 2025 | *Barcelona, Spain*

Recognised Researcher

- Generated 18+ high-fidelity urban flow simulations (≈14 TB each) spanning inlet flow directions from 0° to 340°, supporting AI-based digital twins for fast prediction of air pollutant distribution and odour impact in cities (APPWIND project)
- Resolved cross-platform data interoperability between BSC's in-house solver and the partner university's OpenFOAM environment by converting meshes to OpenFOAM format (ANSA) and exporting velocity, pressure, and turbulent-viscosity fields as VTK (ParaView), troubleshooting node-numbering and partitioning issues to enable the partner's passive-scalar pollutant-dispersion simulations
- Contributed models and simulations within the vCity urban digital-twin platform as part of the platform's modeling team
- Mentored 3 graduate interns and master's students in numerical modeling, CFD workflows, and Python-based data analysis, providing technical guidance on in-house solvers and HPC environments.

Barcelona Supercomputing Center

Sep 2018 – Jul 2024 | *Barcelona, Spain*

First Stage Researcher

- Conducted high-fidelity wall-modeled Large-Eddy Simulations of complex three-dimensional flows involving strong anisotropy, separation, and non-linear turbulent interactions, including NASA Junction flow, NASA Hump, and neutral atmospheric boundary layer flow over urban areas
- Developed a machine-learning-assisted wall model for LES that captured relaminarization (turbulent-to-laminar transition) — a capability physics-based wall models inherently lack — and outperformed the in-house equilibrium wall model in corners and recirculation regions

- Analysed large numerical datasets using Python-based scientific computing and data-analysis workflows
- Performed HPC-based CFD workflows using parallel computing environments, running simulations from under one million to 87 million elements
- Developed reproducible numerical-analysis and post-processing workflows for simulation data interpretation
- Authored a peer-reviewed conference paper as first author (published in Lecture Notes in Computer Science) and contributed to technical publications, conference papers, and presentations in CFD and HPC-related research areas
- Contributed high-fidelity LES data and data-driven wall-modeling research to the HifiTurb project's turbulence-model development
- Provided wall-modeling data and the data-driven wall model used in an Argonne National Laboratory study on XGBoost-based wall modeling in Nek5000; the resulting report cites the first-author conference paper

Indian Space Research Organization (ISRO) *Sep 2008 – Jul 2015 | Thiruvananthapuram, India*

System Engineer

- Served as System Engineer in the Air Breathing Propulsion Project, coordinating design, manufacturing, testing, and validation agencies
- Supported the SCRAMJET engine programme through its evolution from Technology Readiness Level (TRL) 3 to TRL 6, enabling a flight-testable prototype
- Drove ground-combustor testing, instrumentation, and validation activities for propulsion system components
- Covered the full component lifecycle — from engineering drawings (AutoCAD, Autodesk Inventor) through manufacturing, testing, qualification, and final integration
- Coordinated interdisciplinary teams operating under strict quality, reliability, and documentation standards
- Mentored graduate engineering trainees in engineering fundamentals and analytical problem-solving
- Participated in a vehicle characterisation launch campaign, flying dummy engine payloads on a sounding rocket to validate the carrier configuration ahead of the SCRAMJET flight test

PROJECTS

HifiTurb

- Development of high-fidelity LES/DNS data for innovative turbulence models; doctoral research on data-driven wall modeling for LES directly contributed to the project's turbulence-model development, resulting in a first-author conference paper published in Lecture Notes in Computer Science

APPWIND

- Development of high-resolution digital twins for fast prediction of air pollutant distribution and odour impact in cities based on AI applied to CFD models; ran 18+ high-fidelity urban flow simulations (≈14 TB each) covering inlet flow directions from 0° to 340° and generated datasets for reduced-order modeling; coordinated across stakeholders, resolved cross-platform problems, and shared data across the project consortium; continued dataset generation through April 2026

vCity

- Contributed urban flow simulations and dimensionality reduction using Proper Orthogonal Decomposition (POD) to the vCity human-centric urban digital-twin platform. Sampled snapshots every 100 time steps, an interval determined through iterative analysis to extract relevant modes; the dense sampling produced high-fidelity datasets of ≈14 TB per simulation. Also identified a scaling constraint in the parallel QR factorization (LAPACK dorgqr), where the number of processes must keep matrix columns greater than rows to avoid memory errors when increasing partitions

Air Breathing Propulsion Project

- Development of India's SCRAMJET engine — as System Engineer, supported the programme from TRL 3 to TRL 6, coordinating design, manufacturing, testing, and validation agencies and participating in the vehicle characterisation launch campaign ahead of the flight test

PUBLICATIONS & TECHNICAL CONTRIBUTIONS

Radhakrishnan, S., Calafell, J., Miró, A., Font, B., & Lehmkuhl, O. (2024). Data-driven wall modeling for LES involving non-equilibrium boundary layer effects. International Journal of Numerical Methods for Heat & Fluid Flow (Q1 journal), 34(8), 3166-3202.

Radhakrishnan, S., Gyamfi, L. A., Miró, A., Font, B., Calafell, J., & Lehmkuhl, O. (2021). A data-driven wall-shear stress model for LES using gradient boosted decision trees. In International Conference on High Performance Computing, Lecture Notes in Computer Science (pp. 105-121). Cham: Springer International Publishing.

Owen, H., Radhakrishnan, S., Gomez, S., & Lehmkuhl, O. (2019). A Finite-element Viscosity Based Non-slip Wall Modelling Strategy for the Simulation of Industrial Turbulent Flows. ERCOFTAC Bulletin, 120. https://doi.org/10.57651/b120.11

CONFERENCE PRESENTATIONS

International Supercomputing Conference, Online *2021*

Barcelona Supercomputing Center PhD Symposium, Barcelona, Spain *2021*

ParCFD Symposium, Alba, Italy *2022*

2nd Spanish Fluid Mechanics Conference, Barcelona, Spain *2023*

SKILLS

Numerical Modeling & Scientific Computing: Computational Fluid Dynamics (CFD), Large Eddy Simulation (LES), Turbulence Modeling and Wall Modeling, Reduced Order and Surrogate Modeling, Data-driven and AI/ML-assisted Modeling, Scientific workflow automation and post-processing, CFD data analysis and visualization, Verification and Validation (V&V), GMSH, ANSA, Finite Element Method (FEM), Finite Volume Method (FVM)

Programming & Automation: Python (NumPy, SciPy, Pandas, Matplotlib), Fortran, C/C++, Linux-based scientific computing workflows, Scientific scripting and data processing, Matlab, TensorFlow, scikit-learn

High-Performance Computing: HPC environments and parallel workflows, Large-scale simulation data processing, Numerical workflow optimisation

Engineering & Simulation Software: ANSYS (APDL & Mechanical), Finite element-based in-house code development

VOLUNTEERING & OUTREACH

World Space Week (United Nations) — ISRO / Vikram Sarabhai Space Center (VSSC) *Kerala, India | Oct 2010 - Oct 2014*

Volunteer Science Outreach Lecturer

- Delivered outreach lectures to school students in rural areas on rocket science, satellites, and their benefits to humanity
- Participated in the United Nations' World Space Week celebrations, coordinated in India by ISRO and locally by Vikram Sarabhai Space Center (VSSC)
- Communicated complex aerospace topics to non-technical audiences, fostering early interest in science and space exploration

CERTIFICATIONS

Deep Learning Specialization, DeepLearning.AI *Jul 2021*

5-course specialization covering neural network architectures (CNNs, RNNs, LSTMs, Transformers) and training strategies (Dropout, BatchNorm, Xavier/He initialization), applied with Python and TensorFlow to real-world tasks including speech recognition, machine translation, and NLP.

Fundamentals of Parallelism on Intel Architecture, Intel *Aug 2021*

Covers the multiple forms of parallelism on modern Intel architecture processors — data parallelism with vector instructions, task parallelism with threads, distributed-memory parallelism with message passing, and containers — applicable to HPC, datacenter, and AI workloads.

LANGUAGES

English (Fluent), Malayalam (Native), Spanish (Basic), Tamil (Fluent), Hindi (Basic)